Optimizing IT Support Team Performance Using Analytics

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PROJECT OVERVIEW

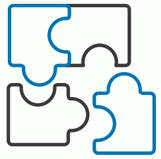
- IT support teams handle a large number of service tickets daily
- Tracking performance and resolution efficiency is challenging
- Delays in resolving issues reduce service quality
- Lack of clear insights affects decision-making
- This project focuses on analyzing ticket data to improve efficiency

DATASET OVERVIEW

- Dataset contains IT service desk ticket information
- Includes details such as priority, category, and ticket status
- Tracks ticket lifecycle from creation to resolution
- Contains resolution time for performance measurement
- Includes customer satisfaction ratings

DATA PREPROCESSING

- Identified and handled missing values
- Removed duplicate and inconsistent records
- Standardized data formats (date, text, categories)
- Transformed raw data into structured format
- Used Power Query for data cleaning and transformation



METHODOLOGY

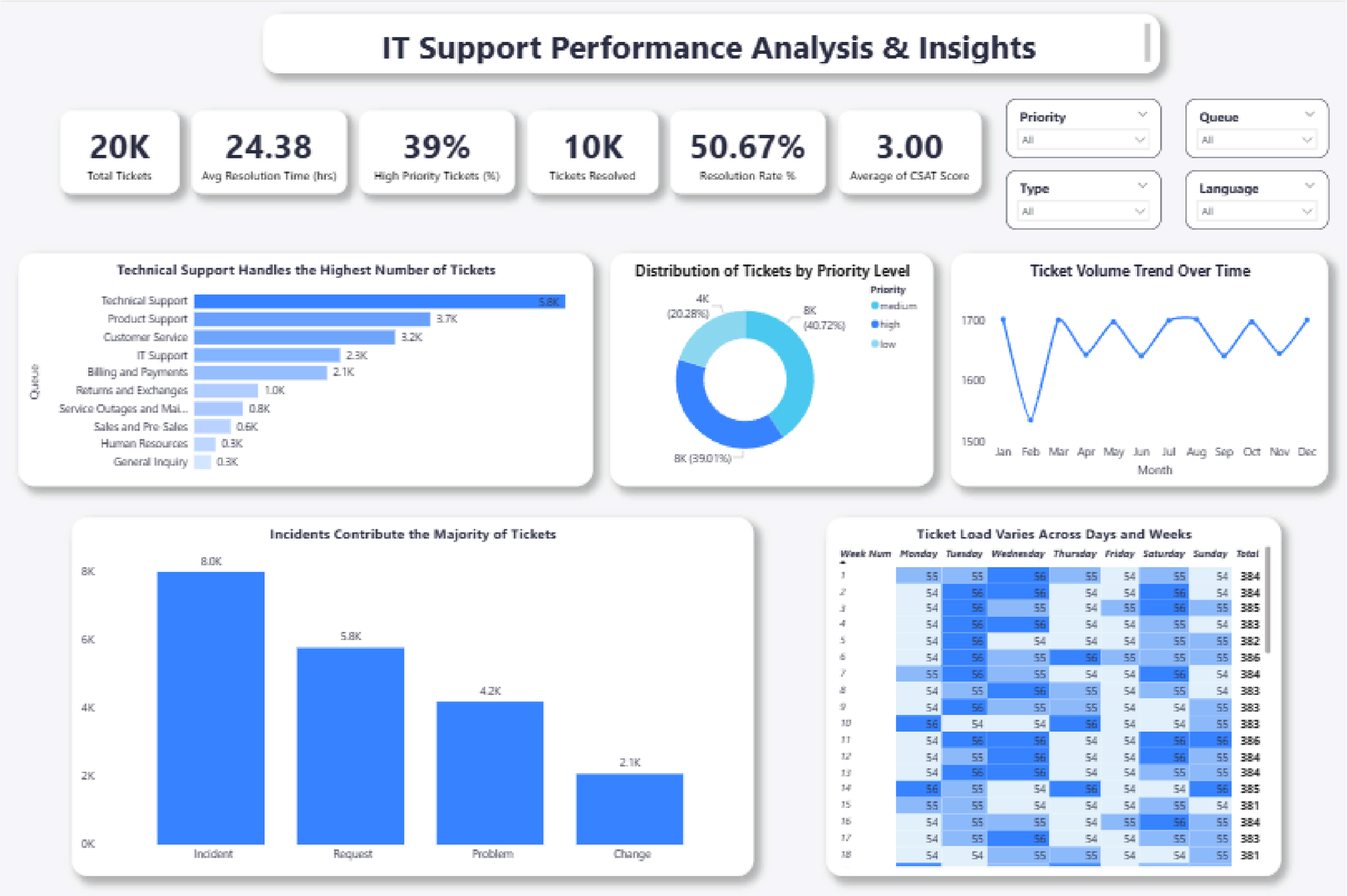
- Collected IT support ticket dataset
- Cleaned and prepared data using Power Query
- Transformed data into structured format
- Created KPIs and measures using DAX
- Built interactive dashboard using Power BI

EXPLORATORY DATA ANALYSIS

- Analyzed ticket distribution across departments
- Studied ticket trends based on priority levels
- Identified most frequent ticket categories
- Observed ticket creation patterns over time
- Compared workload across different periods

VISUALIZATION

- Designed an interactive dashboard using Power BI
- Used bar charts, line charts, and heatmaps
- Implemented slicers for dynamic filtering
- Created KPI cards for quick performance overview
- Maintained clean and consistent design



This Interactive dashboard providing insights into ticket trends, resolution performance, and workload distribution.

KEY INSIGHTS

- Majority of tickets fall under medium priority
- Incident-type tickets have the highest volume
- Ticket volume varies across different time periods
- Peak workload times identified using heatmap
- Resolution rate is approximately 50%

CHALLENGES

- Handling missing and inconsistent data
- Creating accurate KPI calculations using DAX
- Selecting appropriate visuals for data representation
- Maintaining clarity and simplicity in dashboard design
- Ensuring proper filtering and interactivity

FUTURE SCOPE

- Predictive Analytics: Using historical data to predict ticket volumes during peak seasons or system upgrades.
- AI Integration: Implementing AI chatbots for initial ticket screening and automated resolution of common queries.
- Performance Tracking: Expanding the dashboard to track individual agent productivity and training needs.
- Automation: Automating ticket assignment based on the priority and category identified in this study

CONCLUSION

- **Efficiency: This project successfully demonstrates how ticket data analysis can identify bottlenecks in IT support.**

- **Data-Driven Decisions: By visualizing KPIs in Power BI, management can make informed decisions to improve service quality.**

- **Optimization: Improving the resolution rate from 50% will lead to higher customer satisfaction and better resource allocation.**

- **Summary: Overall, the study provides a scalable framework for optimizing IT support performance through analytics**

Thank You